

Enterprise-Grade Power BI Report

Sales Performance & Trend Analysis Dashboard

This document provides a structured overview and step-by-step documentation of the Sales Performance & Trend Analysis Dashboard developed using Power BI Desktop. The content is tailored directly from the assignment requirements and explains what was done in each step and why it was done.

STEP 1: Business Use Case Definition

Objective of the dashboard is to analyze overall sales performance and trends.

Key metrics analyzed include Sales, Profit, and Quantity.

The dashboard enables comparison across time, regions, categories, and products.

Key business questions addressed:

- How are sales trending monthly and yearly?
- Which product categories drive maximum profit?
- Which regions are underperforming?
- How does actual performance compare with targets?

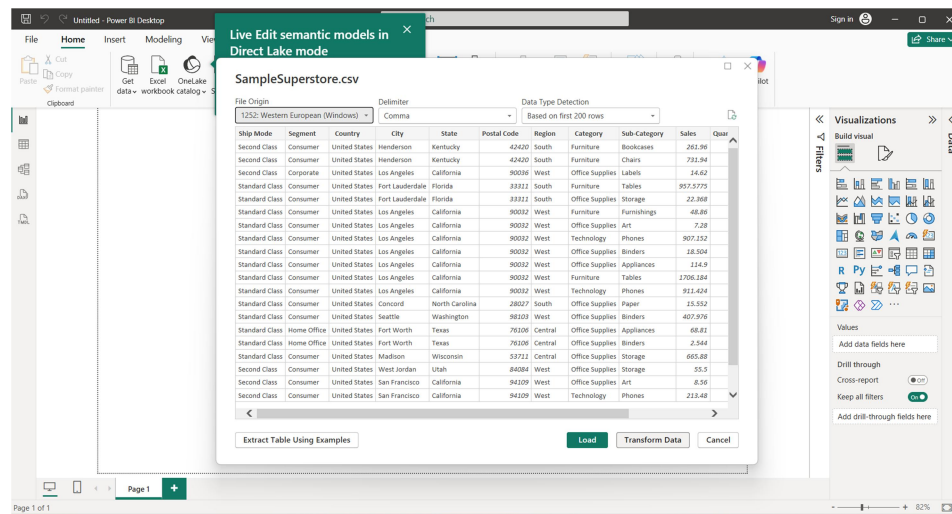
STEP 2: Data Sources & Excel Integration

Excel files were used as the primary data source to simulate real-world business reporting.

Superstore dataset was used as the main sales data source.

A separate Targets.xlsx file was created manually to represent planned sales targets.

Excel was treated as the single source of truth, and Power BI was used as the analytical layer



STEP 3: Power Query Editor – Data Transformation

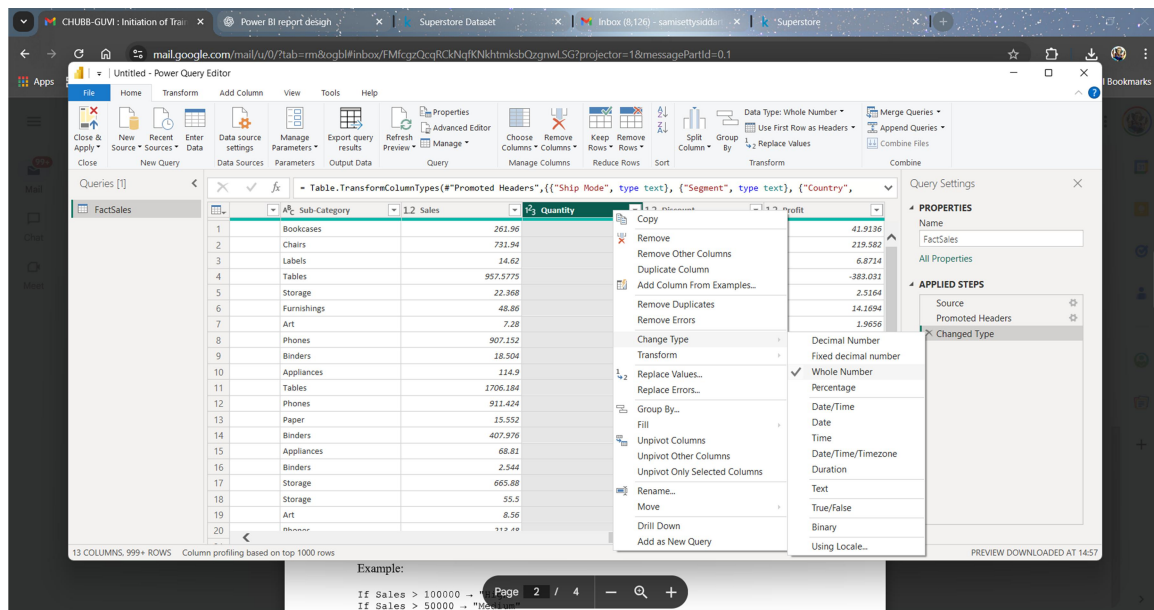
Data cleaning and preparation were performed using Power Query Editor.

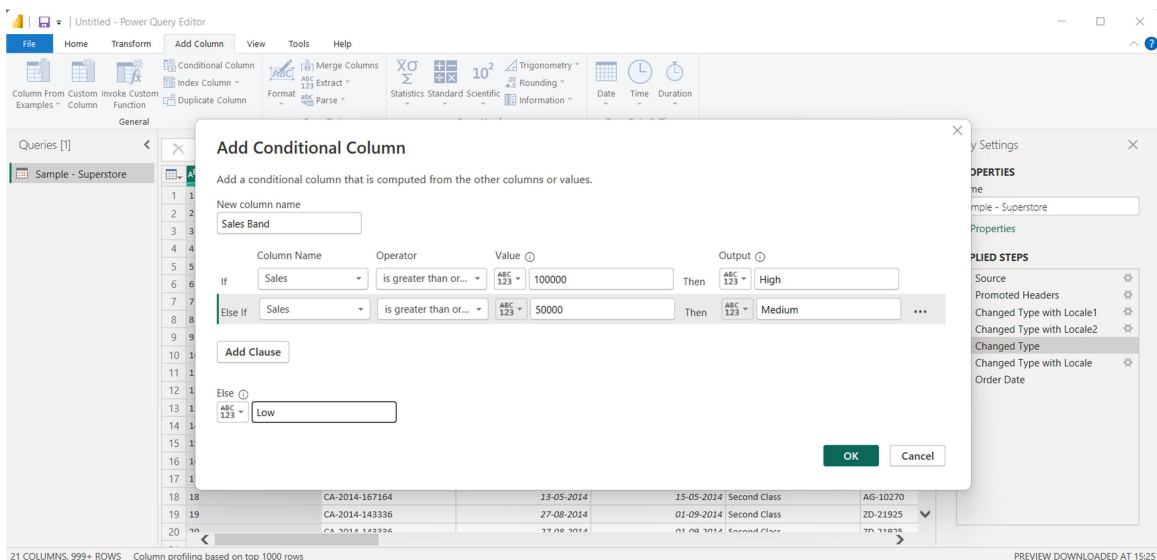
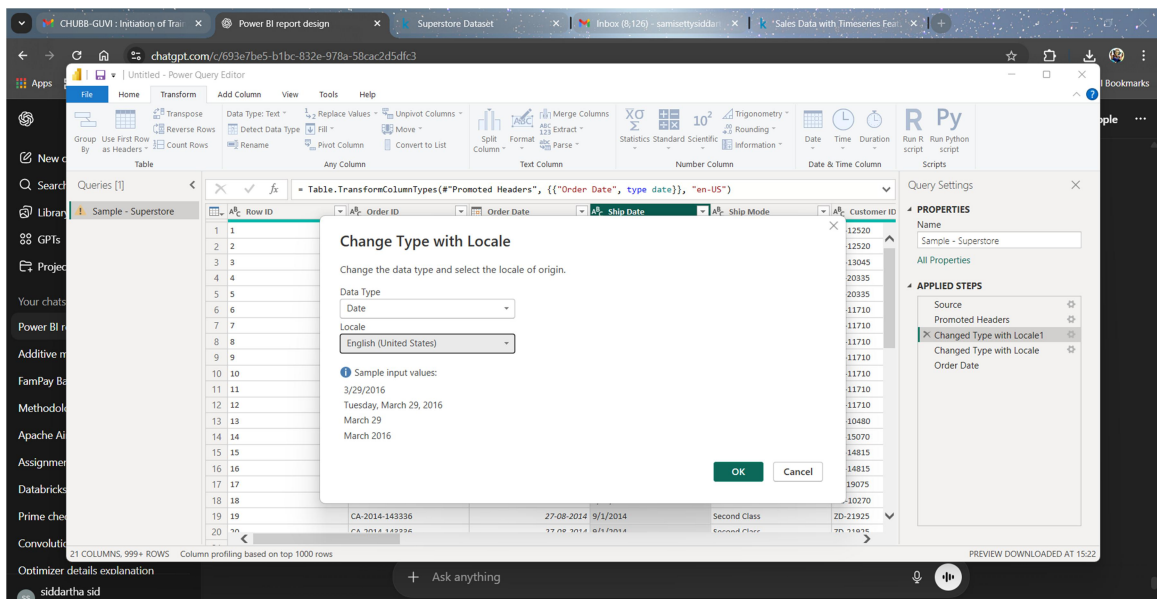
Null values and duplicates were handled to ensure data quality.

Data types were corrected, especially for date and numeric columns.

A conditional column (Sales Band) was created to classify sales as High, Medium, or Low.

These transformations improved performance and ensured consistent business logic.





STEP 4: Data Model Design (Star Schema)

A star schema model was implemented for optimal analytical performance.

FactSales was used as the central fact table.

Dimension tables created: DimDate, DimProduct, DimCustomer, and DimRegion.

Each dimension was deduplicated using a primary key to ensure one-to-many relationships.

DimDate was marked as the official Date table to support time intelligence.

File

Home

Help

Table tools

Manage relationships

Relationships

New measure

measure

Quick measure column

Calculations

New table

Table

Mark as date table

Calendars

Column tools

Name

DimDate

Structure

1

Month Number = MONTH (DimDate[Date])

Date

Year

Month

Month Number

03-01-2014	2014	Jan	1
04-01-2014	2014	Jan	1
05-01-2014	2014	Jan	1
06-01-2014	2014	Jan	1
07-01-2014	2014	Jan	1
08-01-2014	2014	Jan	1
09-01-2014	2014	Jan	1
10-01-2014	2014	Jan	1
11-01-2014	2014	Jan	1
12-01-2014	2014	Jan	1
13-01-2014	2014	Jan	1
14-01-2014	2014	Jan	1
15-01-2014	2014	Jan	1
16-01-2014	2014	Jan	1
17-01-2014	2014	Jan	1
18-01-2014	2014	Jan	1
19-01-2014	2014	Jan	1
20-01-2014	2014	Jan	1
21-01-2014	2014	Jan	1
22-01-2014	2014	Jan	1
23-01-2014	2014	Jan	1
24-01-2014	2014	Jan	1
25-01-2014	2014	Jan	1
26-01-2014	2014	Jan	1
27-01-2014	2014	Jan	1
28-01-2014	2014	Jan	1
29-01-2014	2014	Jan	1
30-01-2014	2014	Jan	1

Table: DimDate (1,458 rows) Column: Month Number (12 distinct values)

File Home Help Table tools

Name DimProduct

Structure

```

1 DimProduct =
2 DISTINCT (
3     SELECTCOLUMNS (
4         FactSales,
5         "ProductID", FactSales[Product ID],
6         "Productname", FactSales[Product Name],
7         "Category", FactSales[Category],
8         "Subcategory", FactSales[Sub-Category]
9     )
10 )
11

```

ProductID ProductName Category SubCategory

OFF-PA-10002377	Adams Telephone Message Book W/Dividers/Space For Phone Numbers, 5 1/4"x8 1/2", 200/Messages	Office Supplies	Paper
OFF-PA-10002005	Xerox 225	Office Supplies	Paper
OFF-PA-10000477	Xerox 1952	Office Supplies	Paper
OFF-PA-10001457	White GlueTop Scratch Pads	Office Supplies	Paper
OFF-PA-10000552	Xerox 200	Office Supplies	Paper
OFF-PA-10000482	Snap-A-Way Black Print Carbonless Ruled Speed Letter, Triplicate	Office Supplies	Paper
OFF-PA-10001725	Xerox 1892	Office Supplies	Paper
OFF-PA-10000477	Xerox 22	Office Supplies	Paper
OFF-PA-10004359	Multicolor Computer Printout Paper	Office Supplies	Paper
OFF-PA-10000254	Xerox 1883	Office Supplies	Paper
OFF-PA-10002893	Wirebound Service Call Books, 5 1/2" x 4"	Office Supplies	Paper
OFF-PA-10004947	Easy-staple paper	Office Supplies	Paper
OFF-PA-10000595	Xerox 1929	Office Supplies	Paper
OFF-PA-10003309	Xerox 211	Office Supplies	Paper
OFF-PA-10001977	Xerox 194	Office Supplies	Paper
OFF-PA-10004782	Xerox 220	Office Supplies	Paper
OFF-PA-10002016	Adams "While You Were Out" Message Pads	Office Supplies	Paper
OFF-PA-10000713	Adams Phone Message Book, 200 Message Capacity, 8 1/16" x 11"	Office Supplies	Paper
OFF-PA-10001526	Xerox 1949	Office Supplies	Paper

Table: DimProduct (1,894 rows)

Data

Search

- DimDate
 - Date
 - Month
 - Month Number
 - Year
- DimProduct
 - Category
 - ProductID
 - ProductName
 - SubCategory
- FactSales
 - Category
 - City
 - Country
 - Customer ID
 - Customer Name
 - Discount
 - Order Date
 - Order ID
 - Postal Code
 - Product ID
 - Product Name
 - Profit

File Home Help Table tools

Name DimCustomer

Structure

```

1 DimCustomer =
2 DISTINCT (
3     SELECTCOLUMNS (
4         FactSales,
5         "Customer ID", FactSales[Customer ID],
6         "Customer Name", FactSales[Customer Name],
7         "Segment", FactSales[Segment]
8     )
9 )
10

```

Customer ID Customer Name Segment

BN-11515	Bradley Nguyen	Consumer
JO-15280	Jas O'Carroll	Consumer
AS-10240	Alan Shonely	Consumer
KD-16345	Katherine Duchich	Consumer
MC-18100	Mick Crebagaga	Consumer
JL-15835	John Lee	Consumer
GT-14755	Guy Thornton	Consumer
XP-21865	Xylona Preis	Consumer
WB-21850	William Brown	Consumer
LS-17230	Lycoris Saunders	Consumer
MV-17485	Mark Van Huff	Consumer
RW-16960	Robert Waldorf	Consumer
CL-12565	Clay Ludtke	Consumer
TB-21250	Tim Brockman	Consumer
FP-14200	Frank Preis	Consumer
RL-18615	Rob Lucas	Consumer
DD-13570	Dorothy Dickinson	Consumer
AT-10735	Annie Thurman	Consumer
JE-15715	Joe Elijah	Consumer
EH-13945	Eric Hoffmann	Consumer

Table: DimCustomer (793 rows)

Data

Search

- DimDate
 - Date
 - Month
 - Month Number
 - Year
- DimProduct
 - Category
 - ProductID
 - ProductName
 - SubCategory
- FactSales
 - Category
 - City
 - Country
 - Customer ID
 - Customer Name
 - Discount
 - Order Date
 - Order ID
 - Postal Code
 - Product ID
 - Product Name

File Home Help Table tools

Name DimRegion

Structure

```

1 DimRegion =
2 DISTINCT (
3     SELECTCOLUMNS (
4         FactSales,
5         "Country", FactSales[Country],
6         "State", FactSales[State],
7         "City", FactSales[City],
8         "Region", FactSales[Region]
9     )
10 )
11

```

Country State City Region

United States	California	Los Angeles	West
United States	California	San Francisco	West
United States	California	San Diego	West
United States	California	Roseville	West
United States	California	Brentwood	West
United States	California	Hesperia	West
United States	California	Costa Mesa	West
United States	California	Santa Ana	West
United States	California	Oceanside	West
United States	California	Encinitas	West
United States	California	Pico Rivera	West
United States	California	Oakland	West
United States	California	San Jose	West
United States	California	Pasadena	West
United States	California	Pomona	West
United States	California	Concord	West
United States	California	Coachella	West
United States	California	La Quinta	West
United States	California	Fresno	West

Table: DimRegion (604 rows)

Data

Search

- DimCustomer
- DimDate
 - Date
 - Month
 - Month Number
 - Year
- DimProduct
 - Category
 - ProductID
 - ProductName
 - SubCategory
- DimRegion
 - Category
 - City
 - Country
 - Customer ID
 - Customer Name
 - Discount
 - Order Date
 - Order ID
 - Postal Code
 - Product ID
- FactSales
 - Category
 - City
 - Country
 - Customer ID
 - Customer Name
 - Discount
 - Order Date
 - Order ID
 - Postal Code
 - Product ID
 - Product Name

← New relationship



Select tables and columns that are related.

From table

DimCustomer

Customer ID	Customer Na...	Segment
BN-11515	Bradley Nguy...	Consumer
JO-15280	Jas O'Carroll	Consumer
AS-10240	Alan Shonely	Consumer

To table

FactSales

Category	City	Country	Customer ID	Customer Na...	Discount	Order Date
Office Supplies	Los Angeles	United States	RA-19885	Ruben Ausman	0	17 July 2017
Office Supplies	Los Angeles	United States	BN-11515	Bradley Nguy...	0	21 Dec 2017
Office Supplies	Los Angeles	United States	BN-11515	Bradley Nguy...	0	21 Dec 2017

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

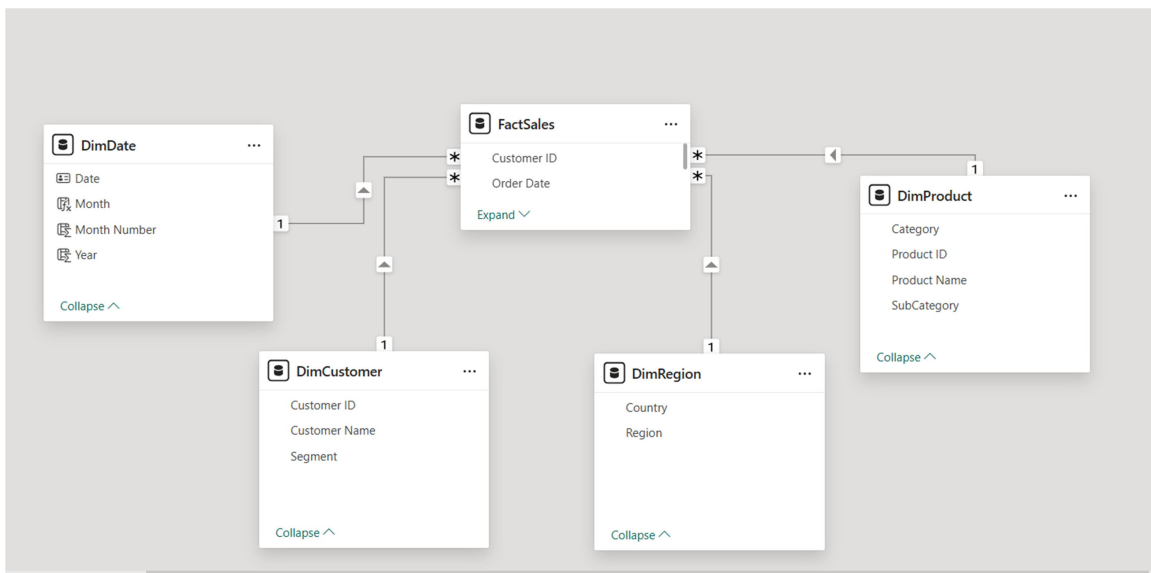
☐ Assume referential integrity

Save

Cancel

+ New relationship Autodetect Edit Delete Filter				
From: table (column)	Relationship	To: table (column)	Status	
☐ FactSales (Customer ID)		DimCustomer (Customer ID)	Active	...
☐ FactSales (Order Date)		DimDate (Date)	Active	...
☐ FactSales (Product ID)		DimProduct (Product ID)	Active	...
☐ FactSales (Region)		DimRegion (Region)	Active	...

Close



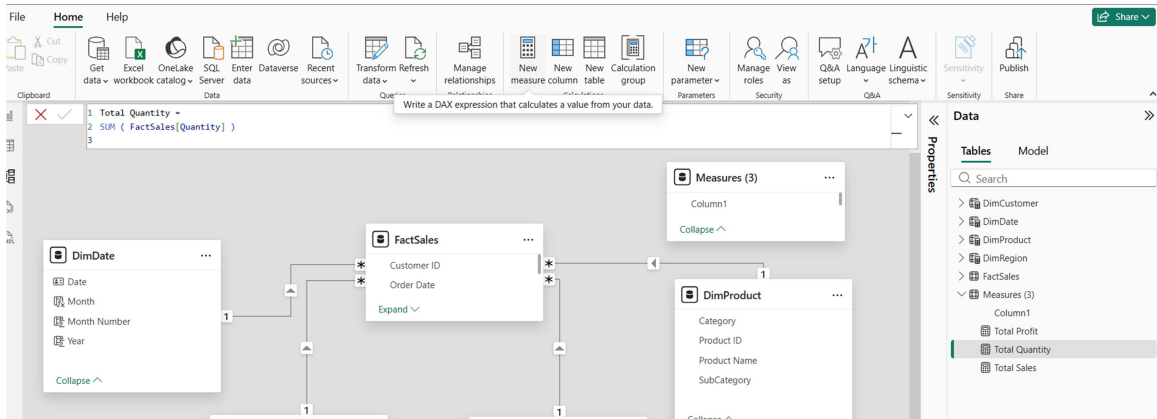
STEP 5: DAX Measures & Calculations

A dedicated Measures table was created to store all DAX calculations.

Core measures such as Total Sales, Total Profit, and Total Quantity were implemented.

Time-intelligence measures like Sales YTD, Sales LY, and YoY Growth % were created.

Measures were used instead of calculated columns for better performance and reusability.



STEP 6: Report Pages & Visual Design

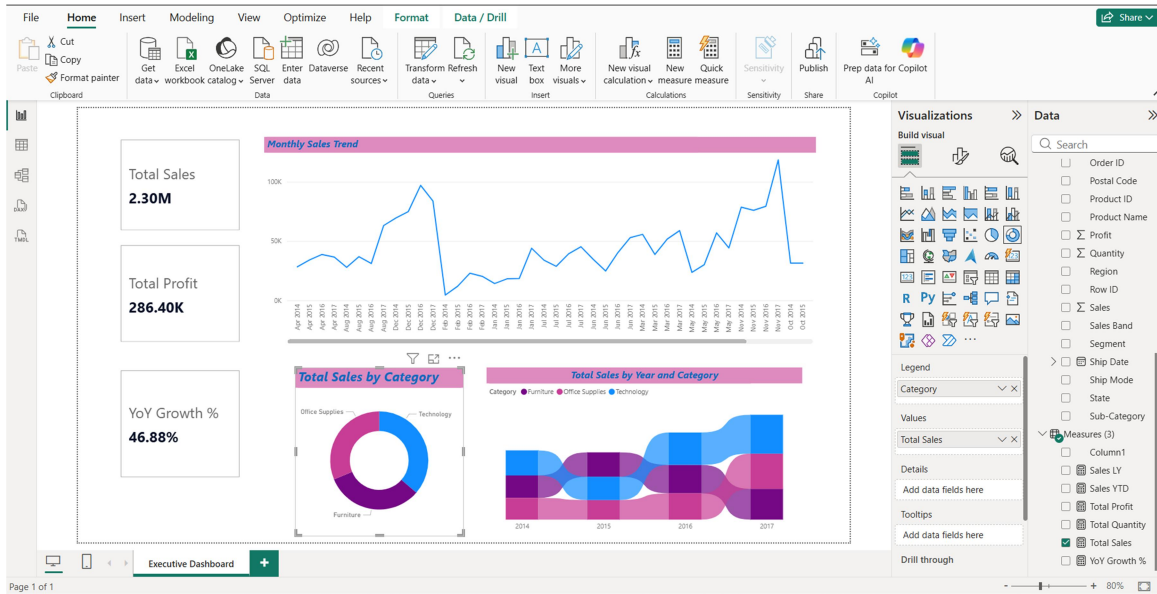
Three report pages were designed to address different analytical needs.

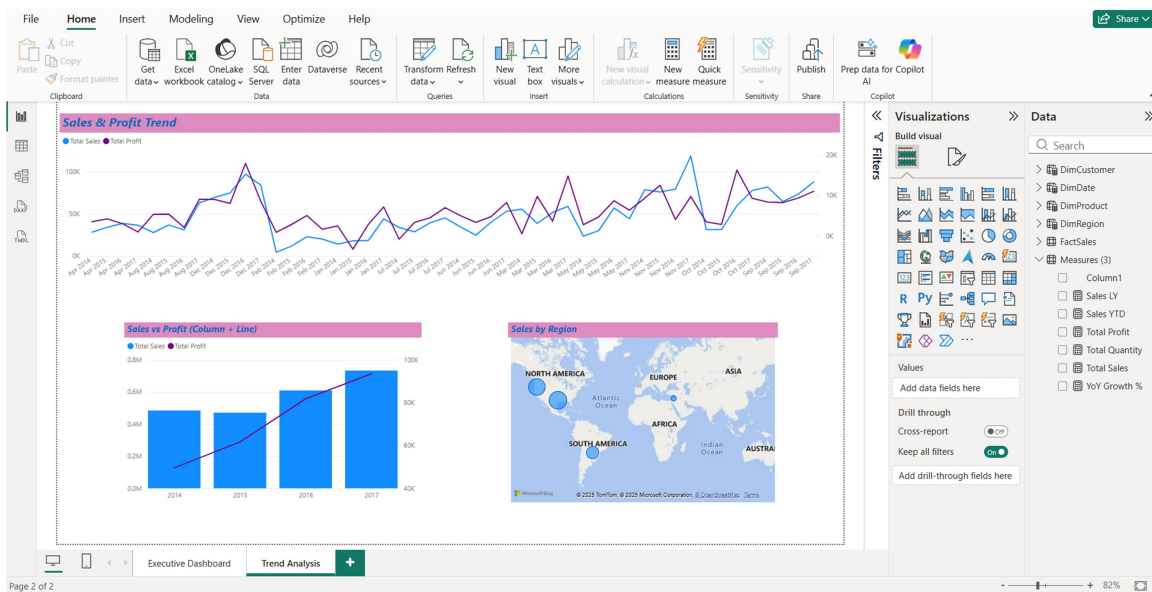
Executive Dashboard page provides a high-level summary using KPI cards and charts.

Trend Analysis page focuses on sales and profit trends over time and across regions.

Product Performance page enables detailed analysis using matrix visuals and drill-downs.

Advanced visuals such as ribbon charts and maps were used for better insights.



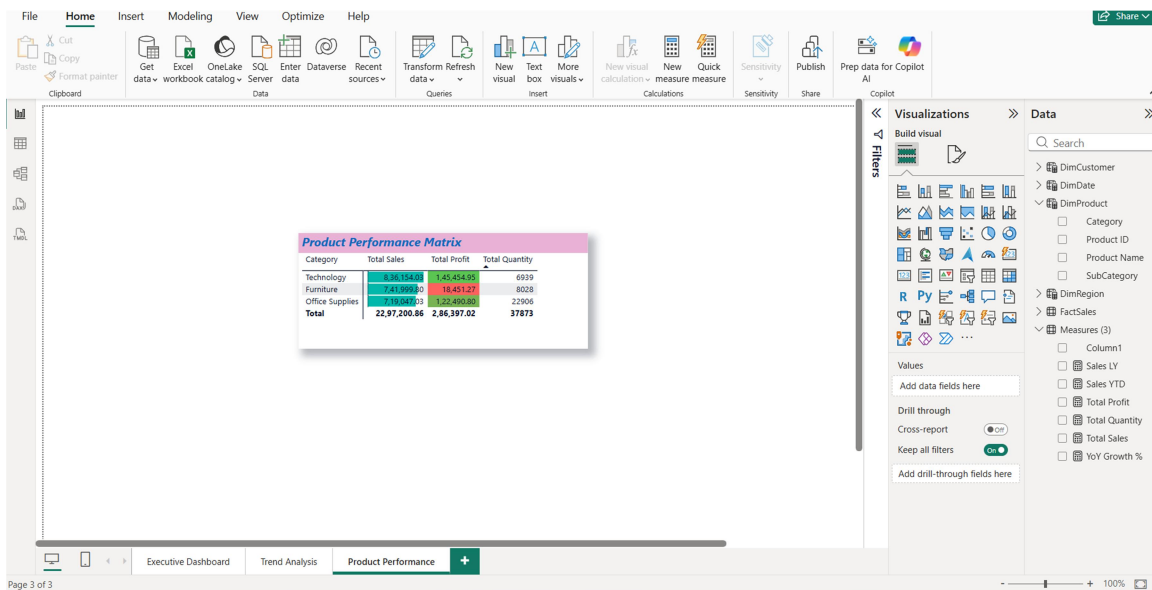


STEP 7: Formatting & Report-Level Filters

Report-level filters for Year, Region, and Category were applied for consistency.

Conditional formatting was used in matrix visuals to highlight performance variations.

Consistent themes, fonts, and layouts were applied across all pages



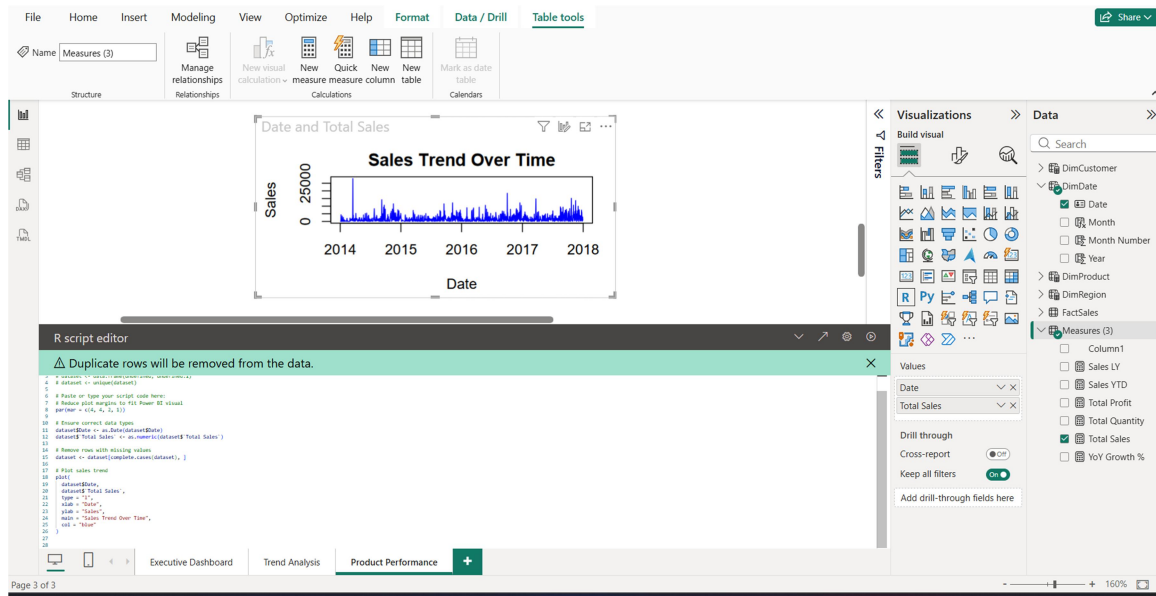
STEP 9: Advanced Visuals – R Script Visual

R was installed and configured to enable advanced analytics in Power BI.

An R Script visual was created to display sales trends over time.

Data type handling and margin adjustments were applied to ensure proper rendering.

This step demonstrates advanced analytical capability beyond standard visuals.



Conclusion

The Sales Performance & Trend Analysis Dashboard was successfully developed using Power BI.

The solution follows enterprise-grade best practices in data modeling and analytics.

It enables interactive, insightful, and scalable business reporting.